Cell Tuition
1-1 PRIVATE TUTORS

GEOMETRY

Angle Rules

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS acute *(koki whāiti)* · obtuse *(koki whānui)* · reflex *(koki hurikōaro)* · straight line *(rārangi torotika)*

A Types of angles ●○○

Remember: Acute: less than 90°. Right angle: 90°. Obtuse: between 90° and 180°. Straight angle: 180°. Reflex: more than 180°.

1. Write the type of each angle.

a) 35° _____ **b)** 90° _____ **c)** 120° _____ **d)** 180° _____

e) 270° _____ **f)** 89° _____ **g)** 91° _____ **h)** 300° _____

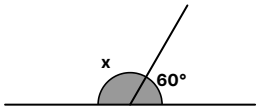
i) 12° _____ **j)** 179° _____ **k)** 181° _____ **l)** 95° _____

B Angles on a straight line ●○○

Remember: Angles on a straight line add to 180°.

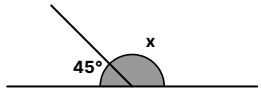
2. Find the size of the angle marked x. Diagrams are not drawn accurately.

a)



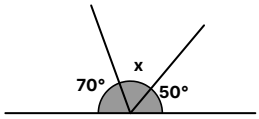
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b)



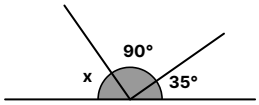
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c)



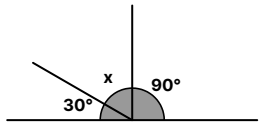
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d)



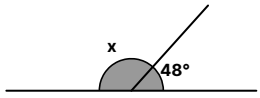
_____

e)



_____

f)



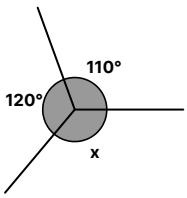
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C Angles at a point ●●○

Remember: Angles that go all the way around a point add to 360°.

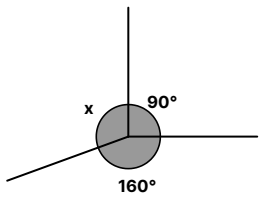
3. Find the size of the angle marked x.

a)



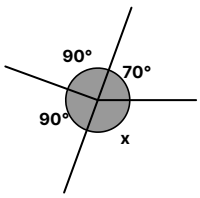
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b)



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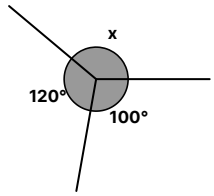
c)



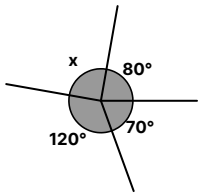
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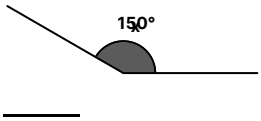
d)



e)



f)

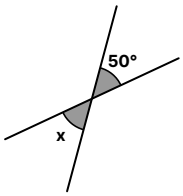


D Vertically opposite angles ●●○

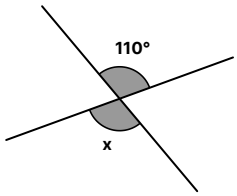
Remember: When two straight lines cross, the angles opposite each other are equal.

4. Find the size of the angle marked x.

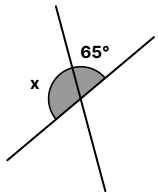
a)



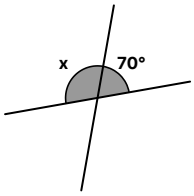
b)



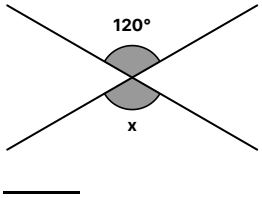
c)



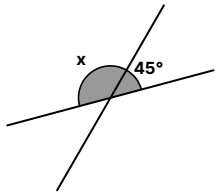
d)



e)



f)

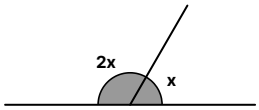


E Using algebra ●●●

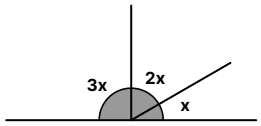
Remember: Write an equation using the angle rule, then solve it.

5. Write an equation and solve it to find x.

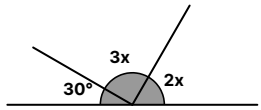
a)



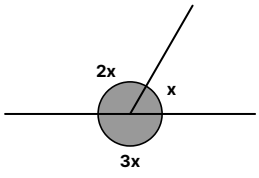
b)



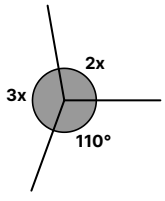
c)



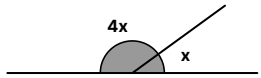
d)



e)



f)



Extension

6. Two straight lines cross. One angle is $(2x + 10)^\circ$ and the angle vertically opposite it is 70°. Find x.

7. Three angles lie on a straight line. One is 55° and the other two are equal. Find the size of each of the equal angles.

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GEOMETRY

Angle Rules

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS acute *(koki whāiti)* · obtuse *(koki whānui)* · reflex *(koki hurikōaro)* · straight line *(rārangi torotika)*

A Types of angles ●○○

Remember: Acute: less than 90°. Right angle: 90°. Obtuse: between 90° and 180°. Straight angle: 180°. Reflex: more than 180°.

1. Write the type of each angle.

a) 50° _____ **b)** 90° _____ **c)** 140° _____ **d)** 180° _____

e) 250° _____ **f)** 88° _____ **g)** 92° _____ **h)** 320° _____

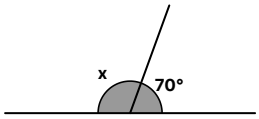
i) 7° _____ **j)** 178° _____ **k)** 182° _____ **l)** 100° _____

B Angles on a straight line ●○○

Remember: Angles on a straight line add to 180°.

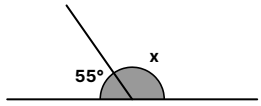
2. Find the size of the angle marked x. Diagrams are not drawn accurately.

a)



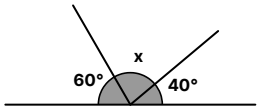
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b)



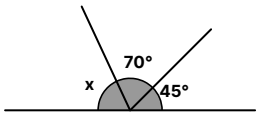
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c)



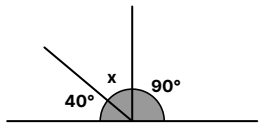
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d)



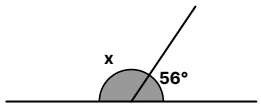
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e)



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f)



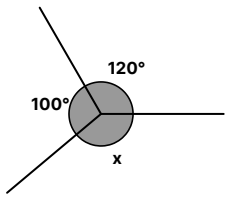
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C Angles at a point ●●○

Remember: Angles that go all the way around a point add to 360°.

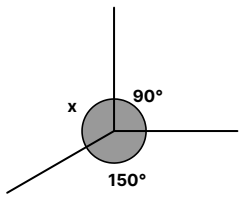
3. Find the size of the angle marked x.

a)



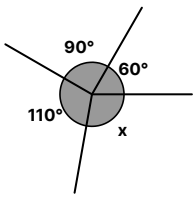
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b)



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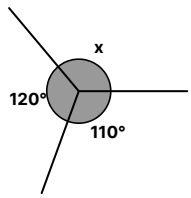
c)



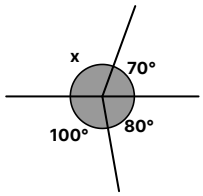
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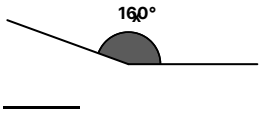
d)



e)



f)

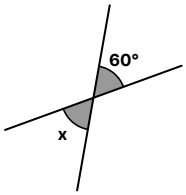


D Vertically opposite angles ●●○

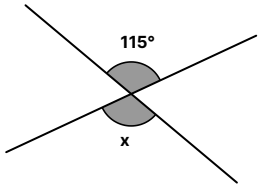
Remember: When two straight lines cross, the angles opposite each other are equal.

4. Find the size of the angle marked x.

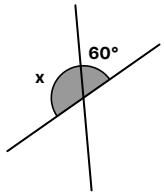
a)



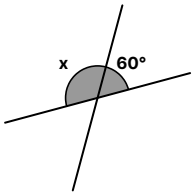
b)



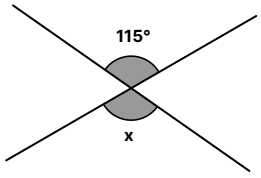
c)



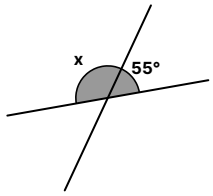
d)



e)



f)

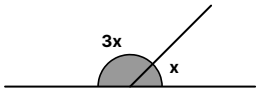


E Using algebra ●●●

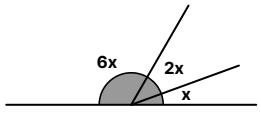
Remember: Write an equation using the angle rule, then solve it.

5. Write an equation and solve it to find x.

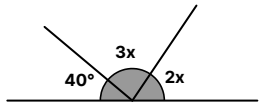
a)



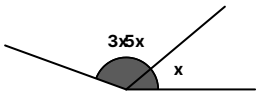
b)



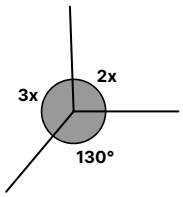
c)



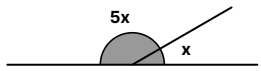
d)



e)



f)



Extension

6. Two straight lines cross. One angle is $(3x + 15)^\circ$ and the angle vertically opposite it is 75°. Find x.

7. Three angles lie on a straight line. One is 65° and the other two are equal. Find the size of each of the equal angles.

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